Matteo Croci | Curriculum Vitae

Oden Institute for Computational Engineering and Sciences, University of Texas at Austin
Peter O'Donnell, Jr. Building, 201 E 24th St, Austin, TX 78712.
matteo.croci@austin.utexas.edu • https://croci.github.io • DOB: 19th June 1992

Research Interests

I am a Research Associate in the Oden Institute of the University of Texas at Austin, working in the groups of Karen E. Willcox and Robert Moser. My research has always been interdisciplinary, working at the interface between different fields in applied mathematics and computational engineering. I work in the fields of computational stochastics, uncertainty quantification and computational and industrial mathematics, with a focus on multilevel (quasi) Monte Carlo methods and biomedical computing. More recently, I have become an expert in reduced- and mixed-precision computing, in particular in the development of reduced- and mixed-precision methods for the numerical solution of differential equations.

My research interests are:

- Multilevel (quasi) Monte Carlo methods.
- Reduced- and mixed-precision numerical algorithms.
- Uncertainty quantification.
- Fast Gaussian field sampling and PDEs with random coefficients.
- Mathematical modelling of the physiology of the brain.
- Finite element method.
- Multiple solutions of variational inequalities.

Current and Previous Research Positions

Research Associate (Research Fellow)
Oden Institute, University of Texas at Austin, Austin, TX, USA — *2022–pres.*
Working in the research groups of Robert Moser and Karen E. Willcox.

Postdoctoral researcher
Oden Institute, University of Texas at Austin, Austin, TX, USA — *2021–2022*
Principal Investigators: Robert Moser and Karen E. Willcox.

Postdoctoral researcher in Computational Stochastics
Mathematical Institute, University of Oxford, UK — *2019–2021*
Principal Investigator: Michael B. Giles.

External PhD student
Simula Research Laboratory, Lysaker, Norway — *2016–2019*

External collaborator
Nielsen, Oxford, UK — *2016*

Education

DPhil (PhD) in Mathematics
University of Oxford, University College, UK, Industrially Focused Mathematical Modelling CDT — *2015–2019*
Thesis: Multilevel Monte Carlo Methods for Uncertainty Quantification in Brain Simulations,
Supervisors: P. E. Farrell, M. B. Giles.
Industrial supervisor: M. E. Rognes (Simula Research Laboratory, Lysaker, Norway).

MSc in Mathematical Modelling and Scientific Computing
University of Oxford, St. Anne's College, UK, final award: ***distinction (highest grade)*** — *2014–2015*
Thesis: Deflation Methods for Complementarity Problems. Published in Optimization Methods and Software.
Supervisor: P. E. Farrell.

BSc in Mathematical Engineering
Politecnico di Milano, Milan, Italy, final award: **110/110 cum laude *(highest grade)*** — *2011–2014*

Exchange Student in Mathematical Engineering
University of Maryland, College Park, MD, USA, academic honors (highest grade) — *2013*

Publications

[1] M. Croci, J. Muñoz-Matute - *Exploiting Kronecker structure in exponential integrators: fast approximation of the action of φ-functions of matrices via quadrature* - to appear in Journal of Computational Science **(Quartile: Q1)** - `arxiv.org/abs/2211.00696` (2022).

[2] M. Croci, G. Rosilho de Souza - *Mixed-precision explicit stabilized Runge-Kutta methods for single-and multi-scale differential equations* - Journal of Computational Physics **(Quartile: Q1)** - `doi.org/10.1016/j.jcp.2022.111349` (2022).

[3] M. Croci, M. B. Giles - *Effects of round-to-nearest and stochastic rounding in the numerical solution of the heat equation in low precision* - IMA Journal of Numerical Analysis **(Quartile: Q1)** - `doi.org/10.1093/imanum/drac012` (2022).

[4] M. Croci, M. Fasi, N. J. Higham, T. Mary, M. Mikaitis - *Stochastic rounding: implementation, error analysis, and applications* - Royal Society Open Science **(Quartile: Q1)** - doi:10.1098/rsos.211631 (2022).

[5] J. Enderlein, D. Sakhapov, I. Gregor, M. Croci, N. Karedla - *Modeling charge separation in charged nanochannels for single-molecule electrometry* - Journal of Chemical Physics **(Quartile: Q1)** - doi:10.1063/5.0074732 (2022).

[6] M. Klöwer, S. Hatfield, M. Croci, P. Düben, T. Palmer - *Fluid simulations accelerated with 16 bit: Approaching 4x speedup on A64FX by squeezing ShallowWaters.jl into Float16* - Journal of Advances in Modeling Earth Systems **(Quartile: Q1)** - doi:10.1029/2021MS002684 (2021).

[7] M. Croci, M. B. Giles, P. E. Farrell - *Multilevel quasi Monte Carlo methods for elliptic PDEs with random field coefficients via fast white noise sampling* - SIAM Journal on Scientific Computing **(Quartile: Q1)** - `arxiv.org/abs/1911.12099` (2021).

[8] M. Croci, V. Vinje and M. E. Rognes - *Fast uncertainty quantification of tracer distribution in the brain interstitial fluid with multilevel and quasi Monte Carlo* - International Journal for Numerical Methods in Biomedical Engineering **(Quartile: Q1)** - doi: 10.1002/cnm.3412 (2020).

[9] M. Croci, P. E. Farrell - *Complexity bounds on supermesh construction for quasi-uniform meshes* - Journal of Computational Physics **(Quartile: Q1)** - doi:10.1016/j.jcp.2020.109459 (2020) .

[10] P. E. Farrell, M. Croci, T. M. Surowiec - *Deflation for semismooth equations* - Optimization Methods and Software **(Quartile: Q1)** - **2020 Charles Broyden prize** - doi:10.1080/10556788.2019.1613655 (2020).

[11] M. Croci, V. Vinje, M. E. Rognes - *Uncertainty quantification of parenchymal tracer distribution using random diffusion and convective velocity fields* - Fluids and Barriers of the CNS **(Quartile: Q1)** - doi:10.1186/s12987-019-0152-7 (2019).

[12] M. Croci, M. B. Giles, M. E. Rognes, P. E. Farrell - *Efficient white noise sampling and coupling for multilevel Monte Carlo with non-nested meshes* - SIAM/ASA Journal on Uncertainty Quantification **(Quartile: Q1)** - doi:10.1137/18M1175239 (2018).

Scholarships, Prizes & Awards

Charles Broyden Prize for the best paper published in OMS in 2020
Optimization Methods and Software (OMS) — *2020*

Oxford-Radcliffe Scholarship
University of Oxford, University College, Oxford, UK — *2015–2019*

Academic honors for the Fall term
University of Maryland, College Park, MD, USA — *2013*

Scholarship for students with very high merit
Politecnico di Milano, Milan, Italy, (awarded every academic year) — *2011–2013*

Award for best freshmen in the academic year
Politecnico di Milano, Milan, Italy — *2011–2012*

Research Stays

Simula Research Laboratory (Norway). Working with Marie E. Rognes.
Duration: 6 months in total across the period. — *2016–2020*

University College London (UK). Visiting John C. Vardakis and Vegard Vinje.
Duration: 2 months. — *2019*

Nielsen (UK). Industrial outreach and research collaboration.
Duration: 3 months. — *2015*
Supervisor: J. Tanner (University of Oxford). Industrial supervisors: G. Mercaldi, L. Daemen (Nielsen, UK).

Invited Seminars Presentations

[IS1] M. Croci, Multi-output multilevel BLUEs via semidefinite programming. **Sandia National Laboratories**, NM, USA. October 14, 2022.

[IS2] M. Croci, A review of mixed-precision methods in scientific computing. Center for Optimization and Statistical Learning seminar. **Northwestern University**, IL, USA. October 6, 2022.

[IS3] M. Croci, Multi-output multilevel BLUEs via semidefinite programming. IMA Data Science seminar. **Institute for Mathematics and its Applications - University of Minnesota**, MN, USA. September 20, 2022.

[IS4] M. Croci, Solving differential equations in reduced and mixed precision. Applied Computational Maths seminar. **University of Minnesota**, MN, USA. September 19, 2022.

[IS5] M. Croci, Mixed-precision order-preserving Runge-Kutta methods. **Exascale computing project** - Multiprecision Effort Team seminar (online). May 26, 2022.

[IS6] M. Croci, Solving differential equations in reduced and mixed precision. Applied Mathematics seminar. **University of Yale**, CT, USA. March 9, 2022.

[IS7] M. Croci, Solving differential equations in reduced and mixed precision. MATHICSE seminar. **École polytechnique Fédérale de Lausanne (EPFL)**, Switzerland. February 2, 2022.

[IS8] M. Croci, M. B. Giles, Solving parabolic PDEs in half precision. Applied Mathematics seminar, **University of Leicester**, UK. March 25, 2021.

[IS9] M. Croci, M. B. Giles, Solving parabolic PDEs in half precision. PEQUAN seminar, **Computer Science Laboratory LIP6 - Sorbonne Université**. February 24, 2021.

[IS10] M. Croci, M. B. Giles, P. E. Farrell, M. E. Rognes, V. Vinje, Non-nested multilevel Monte Carlo methods for PDEs with random coefficients. **Courant Institute of Mathematical Sciences, New York University (NYU)**, NY, USA. February 18, 2021.

[IS11] M. Croci, M. B. Giles, P. E. Farrell, V. Vinje, M. E. Rognes, Non-nested multilevel Monte Carlo methods with applications to brain simulation. Oden Institute Seminar, **University of Texas at Austin**, TX, USA. February 10, 2021.

[IS12] M. Croci, M. B. Giles, Solving (parabolic) PDEs in half precision. Predictability of weather and climate group seminar, **Physics Department - University of Oxford**, UK. January 18, 2021.

[IS13] M. Croci, M. B. Giles, Solving parabolic PDEs in half precision. Numerical linear algebra seminar, **University of Manchester**, UK. December 7, 2020.

[IS14] M. Croci, M. B. Giles, Solving parabolic PDEs in half precision. SCAN department seminar, **Simula Research Laboratory**, Norway. December 2, 2020.

Conference presentations

[C1] M. Croci, G. Rosilho De Souza. *Order-preserving mixed-precision RK methods*. **FEniCS 2022**, La Jolla, CA, USA. August 22-24, 2022.

[C2] M. Croci, M. B. Giles. *Stochastic rounding for parabolic PDEs in half precision*. **SIAM CSE 2021**, virtual conference. March 1-5, 2021.

[C3] M. Croci, M. B. Giles, M. E. Rognes, P. E. Farrell. *MLQMC methods for elliptic PDEs driven by white noise*. **ICIAM 2019**, Valencia, Spain. July 15-March 19, 2019.

[C4] M. Croci, M. B. Giles, M. E. Rognes, P. E. Farrell. *MLQMC methods for elliptic PDEs driven by white noise*. **SIAM CSE 2019**, Spokane, WA, USA. February 25-March 1, 2019.

[C5] M. Croci, M. B. Giles, M. E. Rognes, P. E. Farrell. *Efficient white noise sampling and coupling for multilevel Monte Carlo*. **13th MCQMC**, Rennes, France. July 1-6, 2018.

[C6] M. Croci, M. B. Giles, M. E. Rognes, P. E. Farrell. *White noise coupling for multilevel Monte Carlo*. **SCICADE 2017**, Bath, UK. September 11-15, 2017.

Talks and Participation at Study Groups with Industry

[SG1] M. Croci, S. G. Marquis. Tackling the ExploreSA challenge via machine-learning anomaly detection. Industrial partner: Unearthed (Australia). **MISG 2019**, Adelaide, Australia. January 21-25, 2019.

[SG2] C. Beentjes, D. Chainikov, M. Croci, J. van der Gaast, S. Kapodistria, S. Rahimi-Ghahroodi, F. Sloothaak, F. Spieksma. Optimal order picking from a large retailer warehouse. Industrial partner: CQM (Netherlands). **ESGI 135-SWI 2018**, Eindhoven, Netherlands. January 29-February 2, 2018.

[SG3] M. Croci, P. Morawiecki, J. Prater, V. Sulzer, F. Theil. Classification of Two-Dimensional Gas Chromatography Data. Industrial partner: DSTL (UK). **ESGI 130**, Warwick, UK. September 4-8, 2017.

Organised Conferences and Seminars

Organiser of the North Meet South colloquia
University of Oxford, UK *2020–2021*

Organiser of the Stochastic Simulation seminars
University of Oxford, UK *2019–2021*

Organiser of the FEniCS'18 international conference
University of Oxford, UK *2018*

Supervision and Mentoring

Independent supervision of MSc student M. Robertson (final outcome: distinction)**.**
University of Oxford, UK *2021*
Dissertation title: Mixed-Precision Strong Stability Preserving Runge-Kutta Methods for Hyperbolic Conservation Laws.

Numerical Analysis Group graduate student coordinator
University of Oxford, UK *2018–2021*
Acted as lead mentor to guide and assist all PhD students (≈ 20 at the time) in the Numerical Analysis group

Mentoring of PhD students F. Danieli, K. Kiradjiev, and of MSc student P. Canipet.
University of Oxford, UK *2016–2020*

Participation in Research Projects

[RP1] **US Department of Energy (DOE)**, third Predictive Science Academic Alliance Program (PSAAP III). *Exascale Predictive Simulation of Inductively Coupled Plasma Torches*. Coordinator: Robert D. Moser. Grant number: DE-NA0003969. *2020–2025*

[RP2] **UK Engineering and Physical Sciences Research Council programme (EPSRC)**. *PRISM: Platform for Research In Simulation Methods*. Coordinators: Spencer J. Sherwin, Patrick E. Farrell and others. Grant number: EP/R029423/1. *2018–2023*

[RP3] **UK Engineering and Physical Sciences Research Council programme (EPSRC)**. *Inference, COmputation and Numerics for Insights into Cities (ICONIC)*. Coordinators: Mark Girolami, Nicholas J. Higham, Michael B. Giles, Desmond J. Higham. Grant number: EP/P020720/1. *2017–2019*

[RP4] **European Union's Horizon 2020 research and innovation programme**. *Mathematical and computational foundations for modeling cerebral fluid flow (WATERSCALES)*. Coordinator: Marie E. Rognes. Grant number: 714892. *2017–2023*

[RP5] Project of the **Research Council of Norway**, *The Numerical Waterscape of the Brain (WATERSCAPE)*. Coordinator: Marie E. Rognes. Grant number: #250731. *2016–2020*

[RP6] Project of the **Nordic Council of Ministers via Nordforsk**. *Automated uncertainty quantification for numerical solutions of partial differential equations (AUQ-PDE)*. Coordinators: Joakim Sundnes and Marie E. Rognes. Grant number: #74756. *2015–2018*

[RP7] **UK Engineering and Physical Sciences Research Council programme (EPSRC)** *A new simulation and optimisation platform for marine technology*. Coordinator: Patrick E. Farrell. Grant number: EP/M011151/1. *2015–2017*

[RP8] **UK Engineering and Physical Sciences Research Council programme (EPSRC)** *Automating optimisation subject to partial differential equations on high-performance computers*. Coordinator: Patrick E. Farrell. Grant number: EP/K030930/1. *2013–2018*

Teaching

Tutor in Part A Numerical Analysis
New College, University of Oxford, UK *2021–pres*

Lecturer in Matlab for the MSc in Mathematical Modelling and Scientific Computing
Mathematical Institute, University of Oxford, UK *2020–pres*

Tutor in Stochastic Simulation for the Mathematics of Random Systems CDT
Mathematical Institute, University of Oxford, UK *2019–pres*

Head demonstrator for the Computational Mathematics Course
Mathematical Institute, University of Oxford, UK *2018–2019*

Coordinating all demonstrators, classes and all (around 200) first year students.

Tutor in Advanced Monte Carlo Methods for the MSc in Mathematical and Computational Finance
Mathematical Institute, University of Oxford, UK *2017–2019*

Tutor in Continuous Mathematics
Balliol and Wadham Colleges, University of Oxford, UK *2017–2018*

Demonstrator for the Computational Mathematics course
Mathematical Institute, University of Oxford, UK *2016–2018*

TA in Numerical Analysis for the MSc in Mathematical Modelling and Scientific Computing
Mathematical Institute, University of Oxford, UK *2016–2017*

Tutor for the B6.2 Numerical Solutions of Differential Equations II course
Lincoln College, University of Oxford, UK *2016–2017*

TA in B6.2 Numerical Solutions of Differential Equations II
Mathematical Institute, University of Oxford, UK *2016–2017*

Italian Tutor and Language Partner in the Language Partner Program
University of Maryland, College Park, MD, USA *2013*

Administrative Duties

Early Career Researchers Committee member
Mathematical Institute, University of Oxford, UK *2020–2021*

Health and Safety Committee member
Mathematical Institute, University of Oxford, UK *2020–2021*

Peer-review

I acted as a reviewer for the Mathematical Programming, SIAM Journal on Scientific Computing, Journal of Computational Physics, SIAM Journal on Mathematics of Data Science, and the Quarterly Journal of the Royal Meteorological Society journals.

Membership of Professional Organisations

- **SIAM** - Society for Industrial and Applied Mathematics (Early Career Member).
- **IMA** - Institute of Mathematics and its Applications (Associate Member from 2015 to 2021).

Industrial and Public Outreach Activities

- Presented current research at the InFoMM Annual Meeting at St. Catherine's College (University of Oxford, UK) in front of representatives from a large number of companies, including: Arm, BP, BT, CCFE, Arup, AtdBio, Bbox, DSTL, Schlumberger, JDE, Tesco, Sharp, Siemens, and others. *2017–2020*
- Gave a tutorial on advanced Monte Carlo methods and computational stochastics to end users in the finance industry at the University of Oxford, UK. *2019*
- Helped the organisation of the *It All Adds Up* outreach event at the Mathematical Institute in Oxford whose aim is to inspire young women in year 9-12 to continue with Maths education. *2018*
- Collaborated with the Oxford Mathematical Institute communication staff to publish a short article to disseminate research ideas to the general public (`https://www.maths.ox.ac.uk/node/30078`). *2018*
- Organised a day of outreach lectures aimed at disseminating Maths research ideas to the high school pupils of St Gregory the Great School in Oxford, UK. *2017*
- Visited five companies (McLaren, Sharp, JDE, Dunhumby, Siemens) to promote research collaborations between industry and the Mathematical Institute of the University of Oxford (UK). *2015–2016*

Languages

Italian: native language, **English:** fluent, **Spanish:** fluent, **French:** basic.

Programming Languages & Software

Programming Languages: MATLAB, Python, C++, C, Bash, R.

Software : MPI, OpenMP, CUDA, FEniCS, PETSc, Pybind11, KNITRO, SAS, Git, LaTeX, Gurobi.